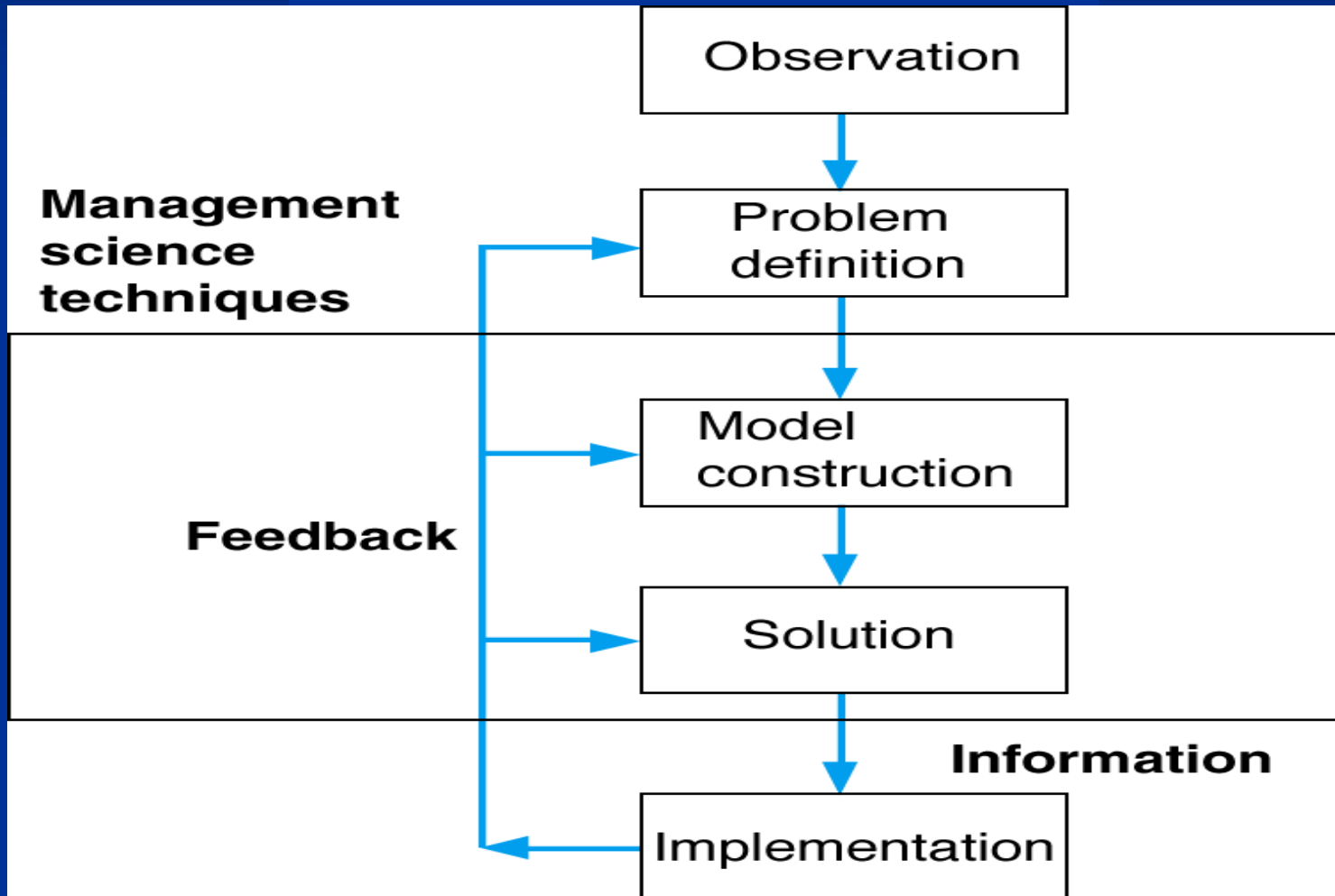


Management Science

Lec. # 1

The Management Science Process

Management science is a scientific approach to solving management problems.



Steps in the Management Science Process

- ✦ **Observation** - Identification of a problem that exists (or may occur soon) in a system or organization.
- ✦ **Definition of the Problem** - problem must be clearly and consistently defined, showing its boundaries and interactions with the objectives of the organization.
- ✦ **Model Construction** - Development of the functional mathematical relationships that describe the decision variables, objective function and constraints of the problem.
- ✦ **Model Solution** - Models solved using management science techniques.
- ✦ **Model Implementation** - Actual use of the model or its solution.

Break-Even Analysis

- ✦ The volume at which **total revenue** equals **total cost** is called the break-even point.
- ✦ Profit at break-even point is zero.

Components of Break-Even Analysis

- The three components of break-even analysis are
 - volume,
 - cost,
 - and profit.

Costs

- Two type of costs are typically fixed costs and variable costs.
- Fixed costs are independent of volume and remain constant.
- Variable costs are determined on a per-unit basis. Thus, **total variable costs** depend on the number of units produced. Variable costs include such items as **raw materials** and **resources**, direct labor, packaging, material handling, and freight.

Model Building: Break-Even Analysis

- **Fixed Costs (c_f)** - costs that remain constant regardless of number of units produced.
- **Variable Cost (c_v)** - unit production cost of product.
- **total variable cost** $= vc_v$
- where c_v = variable cost per unit and v = volume (number of units) sold.
- **Total Cost (TC)** - total fixed cost plus total variable cost.
- **Profit (Z)** - difference between total revenue vp (p = unit price) and total cost, i.e.

$$Z = vp - c_f - vc_v$$

Example #1

As an example, consider Western Clothing Company, which produces denim jeans. The company incurs the following monthly costs to produce denim jeans:

fixed cost = c_f = \$10,000
variable cost = c_v = \$8 per pair

If we arbitrarily let the monthly sales volume, v , equal 400 pairs of denim jeans, the total cost is

$$TC = c_f + vc_v = \$10,000 + (400)(8) = \$13,200$$

The third component in our break-even model is **profit**. Profit is the difference between total revenue and total cost. Total revenue is the volume multiplied by the price per unit,

$$\text{total revenue} = vp$$

where p = price per unit.

Profit is the difference between total revenue (volume multiplied by price) and total cost.

Computing the Break-Even Point

- The break-even point is that volume at which total revenue equals total cost and profit is zero:

$$v = c_f / (p - c_v)$$

(Break-Even Point)

Example #2

For our clothing company example, we have determined total revenue and total cost to be \$9,200 and \$13,200, respectively. With these values, there is no profit but, instead, a loss of \$4,000:

$$\text{total profit} = \text{total revenue} - \text{total cost} = \$9,200 - \$13,200 = -\$4,000$$

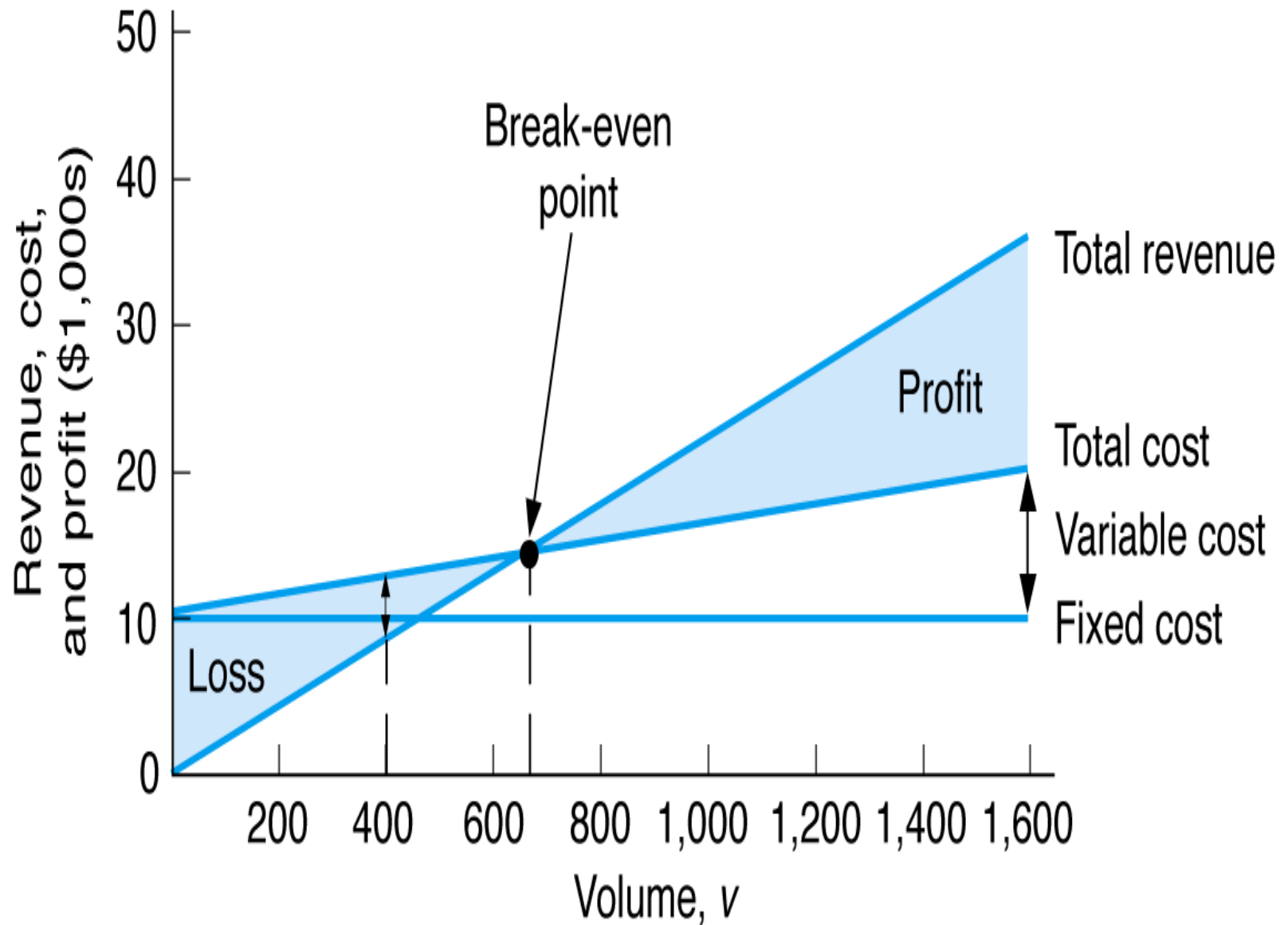
We can verify this result by using our total profit formula,

$$Z = vp - c_f - vc_v$$

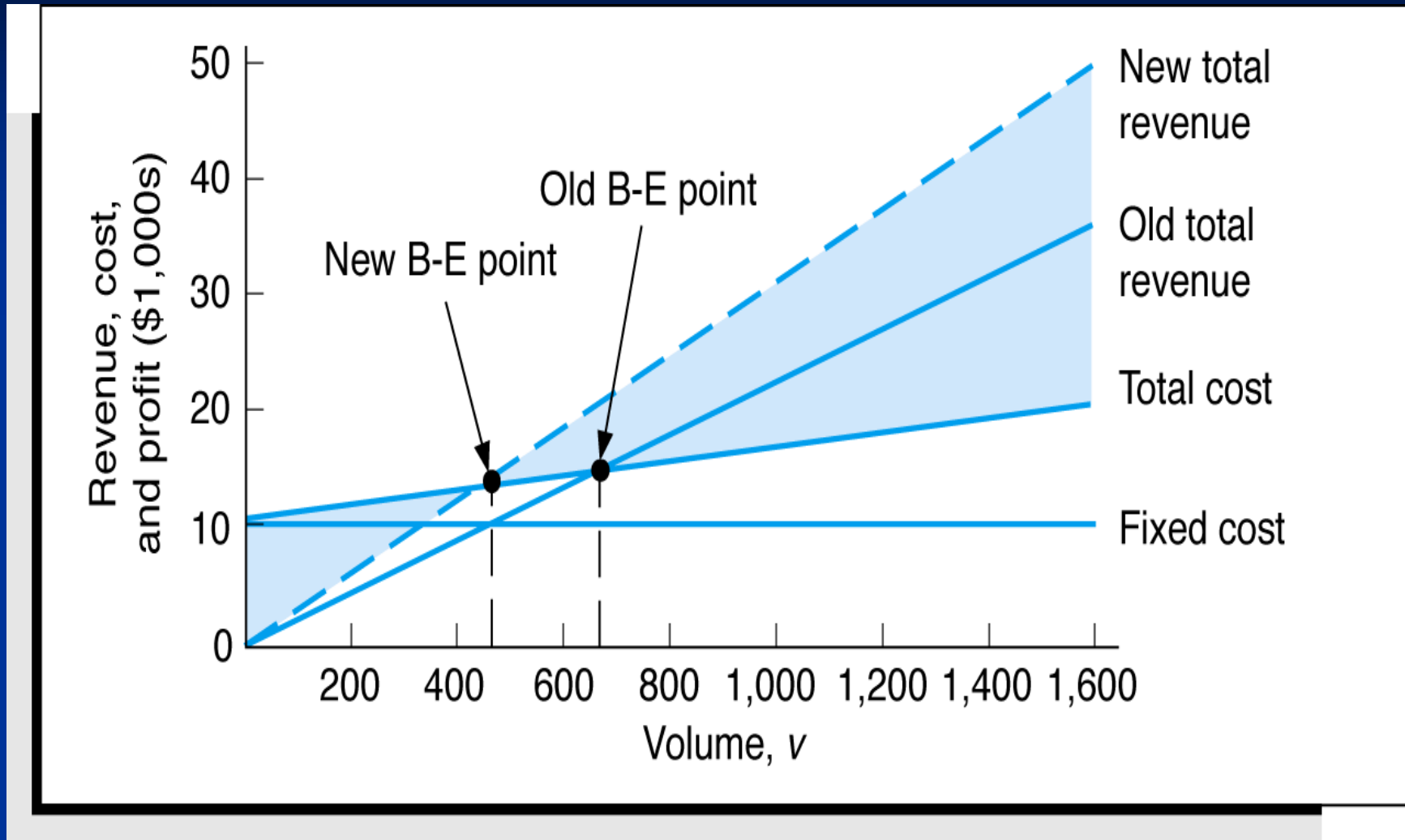
and the values $v = 400$, $p = \$23$, $c_f = \$10,000$, and $c_v = \$8$:

$$\begin{aligned} Z &= vp - c_f - vc_v \\ &= \$ (400)(23) - 10,000 - (400)(8) \\ &= \$9,200 - 10,000 - 3,200 \\ &= -\$4,000 \end{aligned}$$

Break-Even Analysis

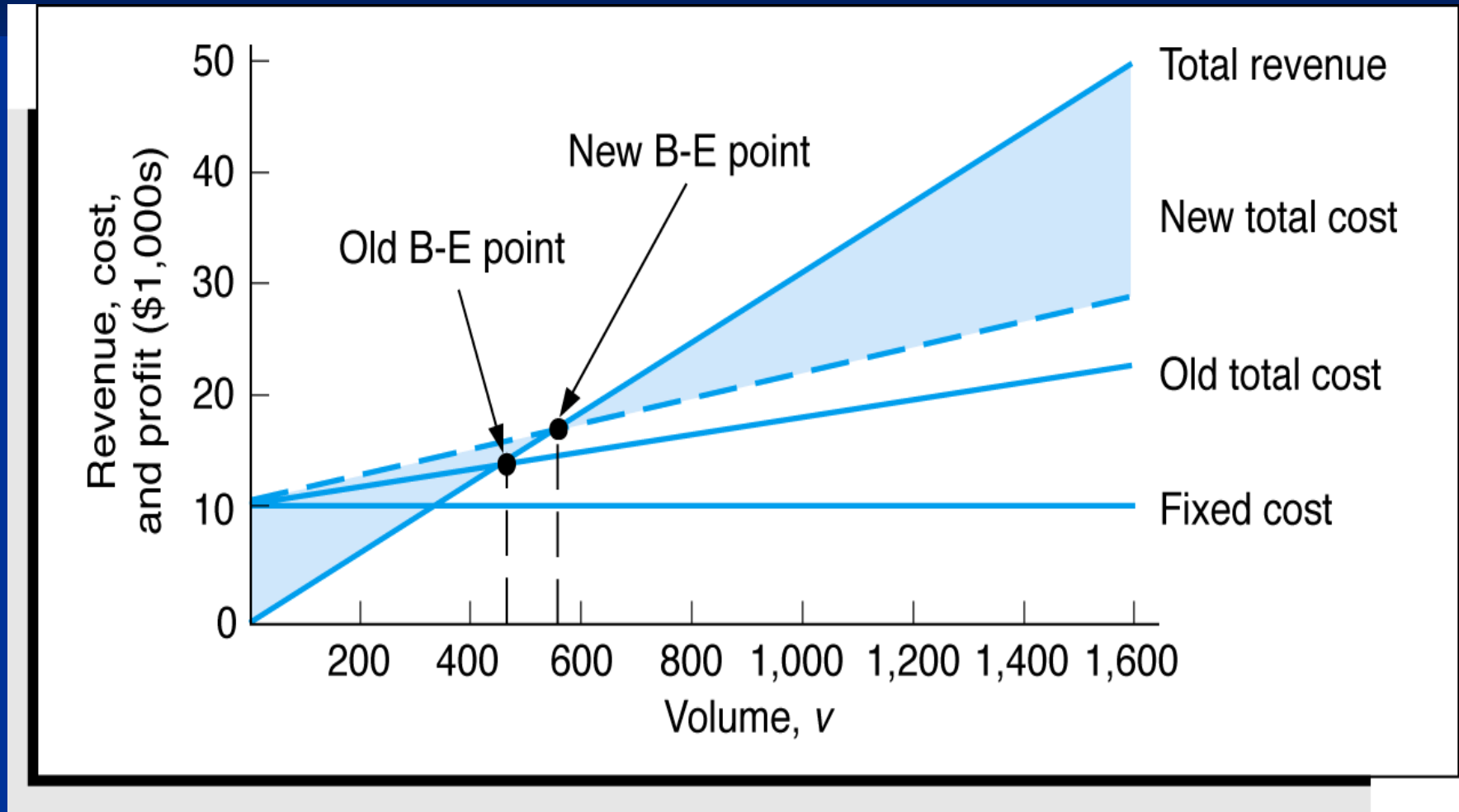


In general, an increase in price lowers the break-even point, all other things held constant.



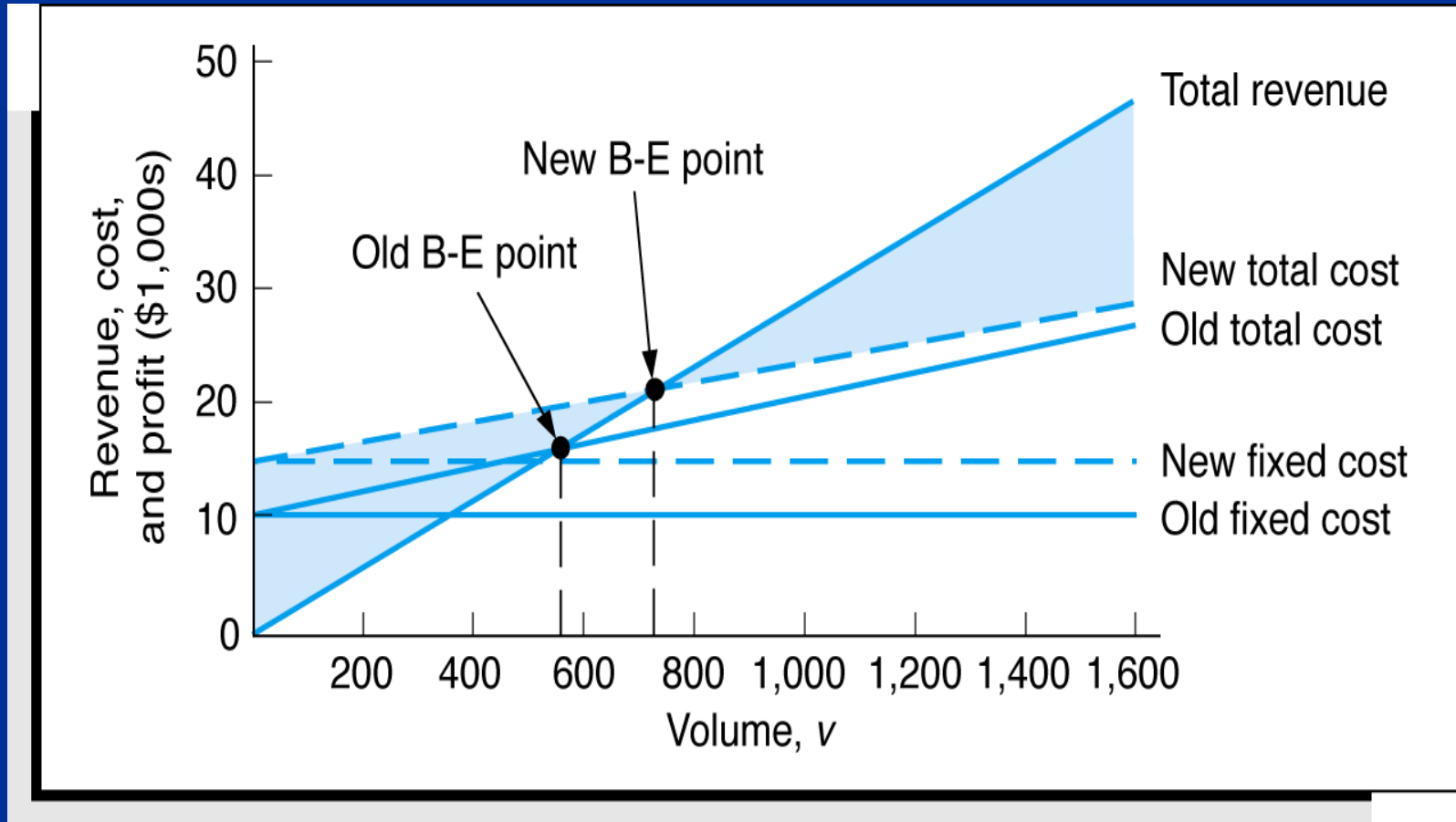
Sensitivity Analysis - Break-even Model with a Change in Price

In general, an increase in variable costs will increase the break-even point, all other things held constant.



Sensitivity Analysis - Break-Even Model with a Change in Variable Cost

In general, an increase in fixed costs will increase the break-even point, all other things held constant.



Sensitivity Analysis - Break-Even Model with a Change in Fixed Cost